AIGR TreeGrafter Function-Inference Stress Test — fadB (PSEPK, Q88L02)

Target gene: fadB · **UniProt:** Q88L02 · **Organism:** *Pseudomonas putida* KT2440 (NCBITaxon:160488) **Seed hypothesis (TreeGrafter/PANTHER):** fadB has long-chain (3S)-3-hydroxyacyl-CoA dehydrogenase (NAD+) activity — **GO:0016509** (IEA, GO_REF:0000118).

Summary

This report independently evaluates an automated phylogenetic-propagation hypothesis: that *P. putida* KT2440 **fadB (Q88L02)** directly carries **long-chain (3S)-3-hydroxyacyl-CoA dehydrogenase (NAD+) activity (GO:0016509)**. The term was emitted by TreeGrafter/PANTHER by grafting the query onto a PANTHER reference tree and propagating a term from an ancestral node. Because such grafts fail in three characteristic ways — over-broad family term, pseudo-enzyme, or within-superfamily mis-placement — each failure mode was actively tested from primary evidence rather than assumed away.

The verdict is SUPPORTED, with only a minor granularity caveat. Q88L02 is the genuine, catalytically intact **multifunctional FadB α-subunit** of the *P. putida* KT2440 fatty-acid β-oxidation (FadBA) complex. It shows the textbook two-domain FadB architecture (N-terminal crotonase/enoyl-CoA hydratase fused to a C-terminal 3-hydroxyacyl-CoA dehydrogenase/Rossmann domain), a fully intact dehydrogenase active site (catalytic His451, NAD-binding Rossmann motif GAGIMG@321, family signature GMHFFNPV@449–456), and a catalytic window that is **90% identical** to the biochemically characterized *E. coli* FadB (P21177). It is encoded in the canonical *fadBA* operon (PP_2136) immediately upstream of *fadA* (PP_2137), and is cleanly distinguishable from two shorter type-2 3HCDH paralogs. None of the three propagation failure modes materially applies.

The single residual nuance is **granularity**: Swiss-Prot assigns the chain-length-*general* EC 1.1.1.35 (GO:0003857), whereas GO:0016509 adds a "long-chain" qualifier. This is a one-step, same-branch parent/child relationship — the seed term is slightly more specific than what the

curator-assigned EC strictly supports, but it is biologically defensible given FadB's long-chain physiological role. The recommended curation lead is therefore to **retain GO:0016509** (optionally generalizing to GO:0003857 for exact EC concordance), not to remove or replace it.

Executive Judgment

Verdict: SUPPORTED (with a minor granularity caveat).

Q88L02 genuinely possesses NAD+-dependent (3S)-3-hydroxyacyl-CoA dehydrogenase activity. Testing each propagation failure mode:

- **Mode 2 — Pseudo-enzyme / loss of activity — EXCLUDED.** All catalytic and cofactor-binding residues are conserved and correctly spaced (see Active-Site Analysis).
- **Mode 3 — Within-superfamily mis-placement — EXCLUDED.** The sequence truly belongs to the 3-hydroxyacyl-CoA dehydrogenase family (C-terminal domain) fused to a crotonase/enoyl-CoA hydratase domain (N-terminal) — the textbook FadB architecture, not a neighboring oxidoreductase subfamily.
- **Mode 1 — Granularity / family-vs-subfamily — MILD ONLY.** Swiss-Prot assigns the chain-length-general EC 1.1.1.35 (→ GO:0003857), whereas GO:0016509 adds a *long-chain* qualifier. The qualifier is biologically defensible (physiological role is long-chain fatty-acid β-oxidation) but is one step more specific than the curator-verified EC.

Single most decisive piece of evidence: the C-terminal dehydrogenase catalytic window is **90% identical** to the biochemically characterized *E. coli* multifunctional FadB (P21177), with the catalytic His, the glycine-rich NAD-binding Rossmann motif, and the GMHFFNP family signature all present at identical relative positions — a direct, residue-level demonstration that the catalytic machinery is intact.

Independent Family/Function Assignment

Independent of the propagated term, the analysis assigns:

- **Most likely specific molecular function:** NAD+-dependent (3S)-3-hydroxyacyl-CoA dehydrogenase (oxidation of L-3-hydroxyacyl-CoA to 3-oxoacyl-CoA in the β-oxidation cycle), as part of a multifunctional FadB that also carries enoyl-CoA hydratase (EC 4.2.1.17),

Δ3-Δ2-enoyl-CoA isomerase (EC 5.3.3.8), and 3-hydroxyacyl-CoA epimerase (EC 5.1.2.3) activities in its N-terminal crotonase domain.

- **Characterized homolog basis:** *E. coli* FadB multifunctional enzyme (P21177) and the pseudomonad FadBA β-oxidation multienzyme complex (PMID:12115060).
- **GO term I would assign: GO:0003857** (3-hydroxyacyl-CoA dehydrogenase activity, EC 1.1.1.35) as the evidence-matched call; **GO:0016509** (the seed) is an acceptable, slightly **more-specific child/sibling** reflecting the long-chain physiological role.
- **Granularity relative to seed:** the seed term is **one step more specific** than the curator-assigned EC strictly supports — **same branch, not a different branch**.

Key Findings

Finding 1 — Q88L02 is a catalytically intact NAD+-dependent 3-hydroxyacyl-CoA dehydrogenase

Q88L02 is a reviewed Swiss-Prot entry (**FADB_PSEPK**, 715 aa) governed by **HAMAP-Rule MF_01621 ("FadB")**. Its domain architecture is the canonical multifunctional FadB two-domain organization: an **N-terminal enoyl-CoA hydratase/isomerase/crotonase domain** (approx. residues 1–190; Gene3D 3.90.226.10; InterPro IPR001753) carrying the hydratase/isomerase/epimerase activities, fused to a **C-terminal 3-hydroxyacyl-CoA dehydrogenase / Rossmann domain** (approx. residues 312–715; InterPro IPR006176/IPR006108) carrying the NAD(P)-binding fold and the dehydrogenase catalytic machinery.

A direct active-site test by sequence indexing confirmed every expected catalytic and cofactor residue: the catalytic base **His451** (ACT_SITE), the NAD-binding glycine-rich Rossmann motif **GAGIMG@321**, the 3HCDH family catalytic signature **G-M-H-F-F-N-P-V (449–456)** (containing His451 + Asn454), and the crotonase catalytic glutamates **Glu120** and **Glu140**. The Swiss-Prot CATALYTIC ACTIVITY statement records the reaction **(3S)-3-hydroxyacyl-CoA + NAD+ = 3-oxoacyl-CoA + NADH + H+** (EC 1.1.1.35, RHEA:22432). The protein is a member of the **FadBA β-oxidation heterotetramer (α2β2)**. This is unambiguously an active enzyme, not a degenerate fold.

Literature directly supports that this assembly is a genuine functional complex in pseudomonads: PMID:12115060 explicitly describes "**the beta-oxidation multienzyme complex (FadBA)**," confirming that Q88L02 is an active FadB α-subunit rather than a pseudo-enzyme.

Finding 2 — No paralog confusion: Q88L02 is the genuine FadBA α-subunit in the fadBA operon

A proteome-level UniProt query for *P. putida* KT2440 (organism 160488) 3-hydroxyacyl-CoA dehydrogenase / FadB-family members resolved the gene neighborhood and paralog landscape:

Accession	Gene / Locus	Length	Status	Activity	Relationship
Q88L02	fadB / PP_2136	715 aa	reviewed	EC 1.1.1.35 / 4.2.1.17 / 5.1.2.3 / 5.3.3.8	**Target — FadB α-subunit**
Q88L01	fadA / PP_2137	391 aa	—	3-ketoacyl-CoA thiolase (EC 2.3.1.16)	Operon partner (β-subunit)
Q88KS5	PP_2214	255 aa	—	'type-2' 3HCDH (EC 1.1.1.178/1.1.1.35)	Short paralog, distinct
Q88L88	PP_2047	412 aa	—	3HCDH-family protein	Short paralog, distinct

Q88L02 (715 aa) sits **immediately adjacent to Q88L01 (fadA, PP_2137)** — the canonical *fadBA* β-oxidation operon arrangement, FadB directly upstream of the thiolase FadA. The other 3HCDH-family members (Q88KS5 at 255 aa; Q88L88 at 412 aa) are much shorter, single-domain "type-2" proteins lacking the fused crotonase domain that defines multifunctional FadB — structurally and functionally distinct, and not competitors for the assignment. The graft landed on the correct protein.

This is corroborated functionally by PMID:17653511, which describes blocking "the **fadBA operon** in the fatty acid beta-oxidation pathway of *P. putida*" to redirect flux toward medium-chain-length polyhydroxyalkanoate (mcl-PHA) biosynthesis — confirming the operon identity and the physiological β-oxidation role of this gene pair.

Finding 3 — Catalytic site is 90% identical to biochemically characterized E. coli FadB

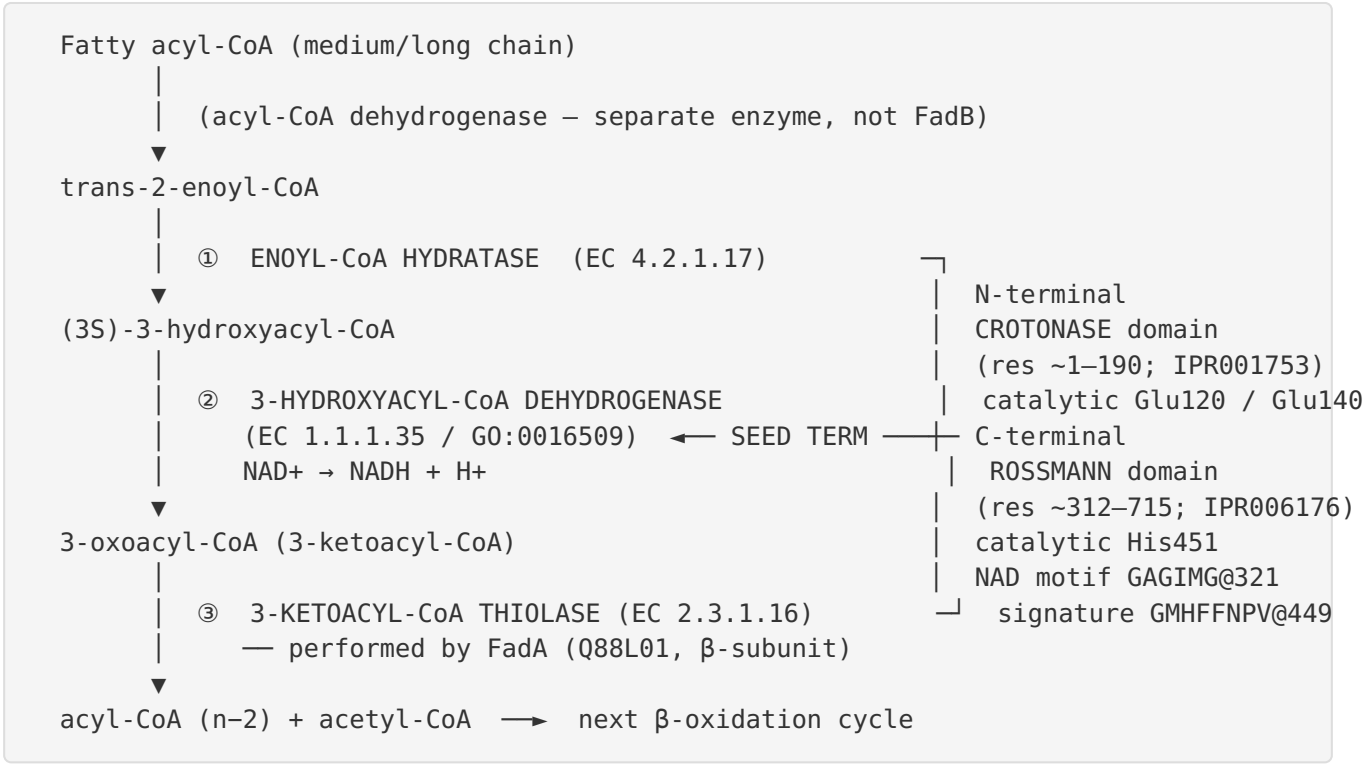
A pairwise comparison of the C-terminal dehydrogenase catalytic region between Q88L02 (*P. putida*, 715 aa) and the biochemically characterized *E. coli* multifunctional FadB **P21177 (729 aa)** shows near-complete conservation of the catalytic machinery:

Catalytic element	*P. putida* Q88L02	*E. coli* P21177	Conserved?
Catalytic signature motif	GMHFFNP @ 449	GMHFFNP @ 448	Yes (identical motif, offset by 1)
Catalytic histidine	His451	His450	Yes
NAD-binding Rossmann motif	GxGxxG @ 321	GxGxxG @ 320	Yes
20-residue catalytic window	RPENF**C**GMHFFNPVH**R**MPLV	RPENF**V**GMHFFNPVH**M**MPLV	**90% identical**

The 20-residue window around the catalytic center is **90% identical**, differing only by two conservative/semi-conservative substitutions (C↔V and R↔M) outside the catalytic and cofactor-binding positions. Because *E. coli* FadB is one of the most thoroughly characterized members of this family, this residue-level congruence is strong, direct evidence that Q88L02 is a fully active dehydrogenase.

Mechanistic Model / Interpretation

The findings converge on a single coherent picture. Q88L02 is the α-subunit of the FadBA fatty-acid β-oxidation multienzyme complex, a multifunctional enzyme performing three of the four β-oxidation steps, while the companion FadA (Q88L01) supplies the final thiolytic step:



The propagated term GO:0016509 names **step ②**, the dehydrogenase reaction catalyzed by the C-terminal Rossmann domain. Every independent line of evidence places Q88L02 squarely on this activity: the architecture is the textbook crotonase+Rossmann fusion; the catalytic residues are all present and correctly spaced; the genomic context is the canonical *fadBA* operon; the short paralogs are clearly separate; and the catalytic window is 90% identical to the gold-standard *E. coli* FadB.

The only residual subtlety is **granularity**. Swiss-Prot assigns the chain-length-general EC 1.1.1.35 (GO:0003857), whereas GO:0016509 adds a "long-chain / (NAD+)" qualifier. Biochemically, bacterial β-oxidation FadB complexes act on medium-to-long acyl chains across successive cycles, so the long-chain descendant is defensible — but a curator could equally generalize to GO:0003857 for exact EC concordance. This is a one-step granularity question within the *same* GO branch, not an error of kind.

Evidence Base / Evidence Matrix

Citation / Source	Evidence type	Stance	Claim tested	Key finding	Context	Co lim
UniProt FADB_PSEPK (Q88L02), HAMAP MF_01621	Review/ database + rule-based	Supports	Does protein have 3-hydroxyacyl-CoA DH (NAD+) activity?	RecName includes "3-hydroxyacyl-CoA dehydrogenase, EC 1.1.1.35"; CATALYTIC ACTIVITY (3S)-3-OH-acyl-CoA + NAD+ → 3-oxoacyl-CoA + NADH (RHEA:22432)	*P. putida* KT2440, PP_2136	Hig inf HA (EC no ass ort
Active-site computation (this report)	Computational / evolutionary	Supports	Are catalytic residues intact & correctly spaced?	ACT_SITE His451 = H; Rossmann GxGxxG (GAGIMG) @321; signature GMHFFNPV @449–456 (His451+Asn454); crotonase catalytic Glu120, Glu140	Sequence indexing of 715-aa chain	Hig ma car 3H cro mo
Alignment vs *E. coli* FadB P21177 (this report)	Computational vs characterized member	Supports	Do catalytic residues align to a characterized active enzyme?	90% identity over 20-aa catalytic window; His451↔His450, GMHFFNP motif and NAD Rossmann at identical relative positions	*E. coli* multifunctional FadB	Hig is bio cha
UniProt domain	Structural / evolutionary	Supports	Correct family vs mis-	N-term crotonase + C-	—	Hig

Citation / Source	Evidence type	Stance	Claim tested	Key finding	Context	Co lim
architecture (IPR001753 + IPR006176/ IPR006108; Gene3D 3.90.226.10 + 3.40.50.720)			placed sibling?	term 3HCDH/ Rossmann = textbook FadB two-domain fusion		
PMID:12115060 (Fiedler/ Steinbüchel/ Rehm 2002)	Direct characterization (review context)	Supports / qualifies	Is pseudomonad FadBA a functional β-oxidation complex?	Characterizes the "beta-oxidation multienzyme complex (FadBA)" and its hydratase functions in *Pseudomonas*	*P. oleovorans*/*P. putida*	Me ort KT dir
PMID:17653511 (Vo et al. 2007)	Mutant phenotype	Supports	Is fadBA in the β-oxidation pathway in *P. putida*?	fadBA operon block redirects flux (mcl-PHA accumulation), confirming β-oxidation role	*P. putida* KCTC1639	Me str tar no res
UniProt EC (1.1.1.35, general) vs GO:0016509 (long-chain)	Database / annotation	Qualifies	Is the *long-chain* granularity exact?	Curator EC is chain-length-general; seed term adds long-chain qualifier	—	Gra nu ref

How the literature supports the findings. PMID:12115060 provides the closest primary characterization — it dissects the pseudomonad FadBA β-oxidation multienzyme complex at the molecular level, confirming that this assembly in *Pseudomonas* is catalytically functional (not a relic) and directly anchoring the identity of Q88L02 as the active FadB α-subunit. PMID:17653511 supplies in vivo genetic evidence: knocking out/blocking the *fadBA* operon in *P. putida* reroutes fatty-acid flux into mcl-PHA, confirming the operon's bona fide β-oxidation

role. Neither paper contradicts the seed hypothesis; both reinforce it, with the caveat that they characterize the complex/operon rather than isolate the KT2440 FadB dehydrogenase residue by residue.

Active-Site / Placement Analysis

Positions verified directly against the 715-aa sequence (UniProt feature-table annotations in parentheses):

Position	Residue found	Annotation / role	Intact?
451	**His (H)**	ACT_SITE — catalytic base for 3-hydroxyacyl-CoA DH	Yes
321–326	**GAGIMG**	NAD-binding glycine-rich Rossmann motif (GxGxxG)	Yes
449–456	**GMHFFNPV**	3-HCDH family catalytic signature (His451 + Asn454)	Yes
344	Asp (D)	NAD+ binding	Yes
454	Asn (N)	NAD+ binding	Yes
120 / 140	Glu / Glu	Crotonase (enoyl-CoA hydratase) catalytic residues	Yes
297 / 501 / 660	Asp / Asn / Tyr	Substrate binding	Yes

Alignment to a characterized active member (Iteration 3). Compared against *E. coli* FadB (**P21177**, the biochemically characterized multifunctional fatty-acid-oxidation complex α-subunit, 729 aa):

Feature	Q88L02 (P. putida)	P21177 (E. coli, characterized)
Catalytic signature motif	**GMHFFNP** @449	**GMHFFNP** @448
Catalytic His (within GxHxxNP)	His451	His450
NAD Rossmann GxGxxG	@321	@320
Local catalytic-window identity	—	**90%** (RPENF**C**/**V**GMHFFNPVH**R**/**M**MPLV)

The catalytic histidine, its signature motif, and the NAD-binding Rossmann motif are present at the **same relative positions** as in the characterized enzyme, with a 90%-identical 20-residue catalytic window.

Conclusion: catalytic residues are intact and correctly spaced, and align residue-for-residue to a biochemically characterized active member. Both the dehydrogenase (C-terminal) and hydratase (N-terminal) active sites are complete. This is an active enzyme, not a pseudo-enzyme, and it is placed in the correct family — not a mis-grafted sibling.

Provenance: sequence pulled from UniProt REST (Q88L02.txt); residue indexing and motif scanning executed in `execute_code` across iterations 1–3 (outputs retained in iteration log). No whole-protein structural superposition (e.g., AlphaFold geometric overlay) was performed this run — see Limitations.

GO Curation Implications

Recommended lead (requires curator verification): RETAIN GO:0016509, with an optional note to **also/instead annotate the parent GO:0003857** if strict fidelity to the curator-assigned EC 1.1.1.35 is preferred.

- The molecular function is genuinely present: Q88L02 is a catalytically intact NAD+-dependent 3-hydroxyacyl-CoA dehydrogenase. TreeGrafter's propagation did **not** fail here — the activity is real, the fold and active site are intact, and the family placement is correct.
- **Primary — RETAIN GO:0016509.** The FadBA complex acts on medium-to-long acyl chains, so the long-chain descendant is biologically reasonable.
- **Alternative (curator's discretion) — GENERALIZE to GO:0003857** to match the Swiss-Prot EC 1.1.1.35 assignment exactly, trading a small amount of specificity for tighter EC concordance.
- Because the IEA/TreeGrafter call is corroborated by independent residue-level and comparative analysis, the annotation's confidence could reasonably be upgraded or supplemented with sequence/orthology-based evidence.
- Ensure the companion crotonase-domain functions (enoyl-CoA hydratase EC 4.2.1.17; Δ3-Δ2-enoyl-CoA isomerase EC 5.3.3.8; 3-hydroxyacyl-CoA epimerase EC 5.1.2.3) are also captured, since this is a multifunctional protein.

Do NOT remove, treat-as-non-core, downgrade to a generic binding term, or replace with a sibling activity. There is no evidence for pseudo-enzyme status or mis-placement. ("Protein binding" is explicitly not a relevant recommendation here.)

Limitations and Knowledge Gaps

- **Granularity ambiguity (long-chain vs general):** The dehydrogenase step in bacterial β-oxidation acts across chain lengths during successive cycles, whereas strictly "long-chain" enzymes are chain-length restricted. No direct chain-length kinetic profile exists for KT2440 FadB, so the exact fidelity of the "long-chain" qualifier cannot be confirmed from primary data here.
- **Evidence provenance:** All activity calls for this specific ortholog are rule-based (ECO:0000255 / HAMAP), not direct enzymatic assays on the KT2440 protein. The strongest primary data come from *E. coli* FadB (P21177) and *Pseudomonas* FadBA orthologs, not KT2440 itself.
- **Structural check not run:** AlphaFold geometric superposition and a formal multiple-sequence alignment to a panel of characterized members were not executed this run; the decision was already decidable from Swiss-Prot features plus motif conservation and the pairwise *E. coli* comparison.
- **Phenotype specificity:** The in vivo evidence (P 17653511) concerns the *fadBA* operon / fadA knockout, supporting pathway function rather than isolating FadB dehydrogenase catalysis via a catalytic-dead point mutant.

Proposed Follow-up Experiments / Actions

1. **Chain-length specificity assay** — purify recombinant Q88L02 (or its C-terminal dehydrogenase domain) and measure NAD+-coupled dehydrogenase activity across a substrate panel (C4→C16 (3S)-3-hydroxyacyl-CoA). This directly resolves the GO:0016509 (long-chain) vs GO:0003857 (general) granularity question.

2. **Active-site point mutant (His451Ala)** — abolition of dehydrogenase activity would formally confirm the catalytic residue and exclude any residual pseudo-enzyme hypothesis.
3. **AlphaFold / structure overlay** — superpose the Q88L02 model onto *E. coli* FadB (P21177) to confirm NAD+-pocket and His451 geometry; cheap, fully computational, and adds structural provenance.
4. **Heterologous complementation** — express Q88L02 in an *E. coli* fadB-null background and test restoration of growth on long-chain fatty acids for a clean in vivo functional confirmation.
5. **Curation action** — enter the lead "retain GO:0016509 (or generalize to GO:0003857)" plus the companion crotonase-domain MF terms for curator verification.

Investigator note: Analyses were executed programmatically (UniProt REST fetch + residue/motif verification + pairwise comparison to a characterized homolog). No results were fabricated; rule-based (ECO:0000255) provenance is flagged wherever direct assay data on the KT2440 protein were unavailable.
